

# ZITENG YANG

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## EDUCATION

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**Georgia Institute of Technology (Georgia Tech)**, Atlanta, GA, USA 2021.08 - 2027.03

*Ph.D. candidate* in Computer Science, advised by Dr. Vivek Sarkar

- **Thesis & Research Areas:** Formal Verification, Compilers
- **Graduate Courses:** Advanced Algorithms (convex optimization and sampling), High-Performance Computer Architecture (multi-core CPU simulator), Compiler Design (LLVM), Parallelizing Compiler, Software Analysis and Testing, [Minor] Advanced Graph Theory, Measure Theory (= Real Analysis)

**Shanghai Jiao Tong University (SJTU)**, Shanghai, China 2017.09 – 2021.06

*B.E.* in Computer Science, with GPA=86; minor in Music

- **Thesis & Research Areas:** Math. Logic (first-order & modal), Program (Separation) Logic & Semantics
- **Selected Courses:** Theory of Computation (Turing, Gödel), Programming Language Theory (Typed  $\lambda$ ), Machine Learning (NN, RL), Linux Kernel (memory pages, process/file system), Computer Network, Information Security, Abstract Algebra, Circuit Theory, Digital Electronics, Mechanics Theory, Game Design, Music Theory, Music Composition, Solfeggio, etc.

## INDUSTRIAL EXPERIENCE

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**Amazon Web Services, Inc. (AWS)**, Automated Reasoning Group & Annapurna Labs

Formal verification of tensor compiler (*AWS Neuron*) code base assisted by theorem prover (*Lean*);

*Applied Scientist Internship*, Seattle, WA, USA

Sep. ~ Dec. 2026

*Applied Scientist Internship*, Seattle, WA, USA

Jan. ~ May. 2025

- Formal verification of internal compiler codebase assisted by Lean 4 prover and Z3 SMT solver

**Meta Platforms, Inc.**, PyTorch & Triton Team

*Compiler Engineer Internship*, Menlo Park, CA, USA

Jun. ~ Aug. 2026

*Compiler Engineer Internship*, Bellevue, WA, USA

May. ~ Aug. 2025

Research & Engineering in tensor compiler (*Triton* open-source language, based on LLVM/MLIR)

- Static & sound bitwise-equivalence checker of tensor IR and CUDA PTX / AMD GCN assembly
- Data-layout optimization for inner tree reduction
- Triton general matrix multiply (GEMM) GPU kernels bit-matching NVIDIA cuBLAS (closed source) with PyTorch Inductor integration and cuBLAS tensor dropping bug reproduced (AI agent workflow assisted)
- Debugging information preservation support in Triton's compiler

## RESEARCH PAPERS

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**Z. Yang**, N. Riasanovsky, W. Deng, V. Sarkar, Taming Bitwise Behavior in GPU Kernel and Compiler (in submission)

**Z. Yang**, J. Shirako, V. Sarkar, Fully Verified Instruction Scheduling, *OOPSLA'24*

**Z. Yang**, X. Yin and S. Li, Maximally permissive supervisor control of timed discrete-event systems under partial observation, in *21st IFAC World Congress, 2020*

## RESEARCH EXPERIENCE

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**Doctorate Thesis: Compiler Correctness for CPU and GPU Language** Aug. 2021- May. 2027

Advisor: Dr. Vivek Sarkar, School of Computer Science, Georgia Tech.

- **[Full Correttness] Fully Verified Instruction Scheduling**  
[OOPSLA'24] Built the *first ever* foundational formal theory to fully verify instruction scheduling passes in verified CPU compiler and achieved the first fully verified scheduling pass in proof assistant (Rocq) .

- [Posterior Validated Correctness] Translation Validation of GPU Kernel Compiler
- [Emperical Correctness] Taming Bitwise Behavior in GPU Kernel and Compiler

## Bachelor Thesis on Program Logic, Proof Assistant and Formal Semantics

2020 – 2021

Advisor: Dr. *Qinxiang Cao*, John Hopcroft Center for Computer Science, SJTU.

- **Semantics Preservation for Annotation Verifier:** Designed and formalized (in Rocq) an extended semantics preservation framework for optimizing verified programs by exploiting information in Hoare assertions
- **Formalized Modal Logic:** Practice on Rocq by formally proving completeness theorem of Modal Logic.
- **Mathematical Logic Essentials:** Mathematical logic (First-order/Modal Logic) with soundness/completeness proof, and Hoare/(Concurrent) Separation Logic with soundness proof

## Formal Control Theory of Timed Discrete-Event Systems

Summer 2019

Advisor: Dr. *Xiang Yin*, Department of Automation, SJTU.

- [IFAC'20] Designed a synthesizing algorithm of safe and maximally permissive supervisor for Timed Discrete Event System and proved the language generated by the automata under the synthesized supervisor is safe yet maximal.

## ACADEMIC SERVICES

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CGO 2027 Artifact Evaluation Committee

PLDI 2025 Artifact Evaluation Committee (Reviewed 3 artifacts)

## TEACHING EXPERIENCE

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**Teaching Assistant**, CS6390 Programming Languages, Georgia Tech, by *Vivek Sarkar* Spring 2023

**Teaching Assistant**, CS4510 Automata and Complexity, Georgia Tech, by *Joseph Jaeger* Fall 2022

**Teaching Assistant**, MA208 Discrete Mathematics (IEEE Honor Class), SJTU, by *Qinxiang Cao* Fall 2020

**Teaching Assistant**, MA239 Discrete Mathematics (Zhiyuan Honor Class), SJTU, by *Xiang Yin* Fall 2020

## SKILLS

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### Programming Lanugages & Software Tools:

- **C/C++, Python, Java:** professionally skilled in industrial development and academic research
- **Lean 4, Rocq (previously Coq), OCaml, Z3 (SMT Solver):** long-term research on formal verification
- **Tools / Libraries:** LLVM / MLIR (compiler IR tool-chain), Triton (tensor language & compiler), CUDA (Nvidia's GPU language), GDB& LLDB& CUDA-GDB (debugging tools), CompCert (a fully certified C compiler), Iris/VST (separating logic tools for verification), Linux Kernel Programming
- **Unity 5, Unreal Engine:** course projects & student org. experience on video game design
- **Logic Pro, Sibelius:** course projects experience on music composition

**Non-Tech:** Tennis, Piano

## HONORS AND SCHOLARSHIPS

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- Rongchang (荣昶) Scholarship for Science and Technology Innovation, Finalist, 10,000 CNY (20 finalists/winners among university) 2020
- Undergraduate Excellence Scholarship, 500 CNY 2018
- 1st Prize (50 winners per province) in National High School Mathematics League 2016